



Vector

The Accuracy Tracking System

Our Team:



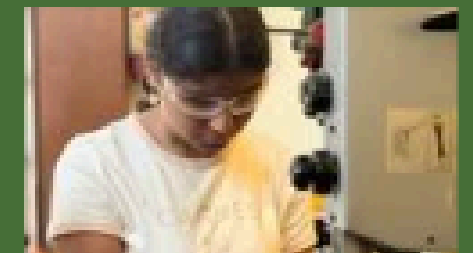
Vidyut Sundar: Plays Tennis Nationally,
an avid interest in Sports Tech



Anjali Randery: Great interest in
Badminton



Saanvi Singh: Enjoys Tennis as well

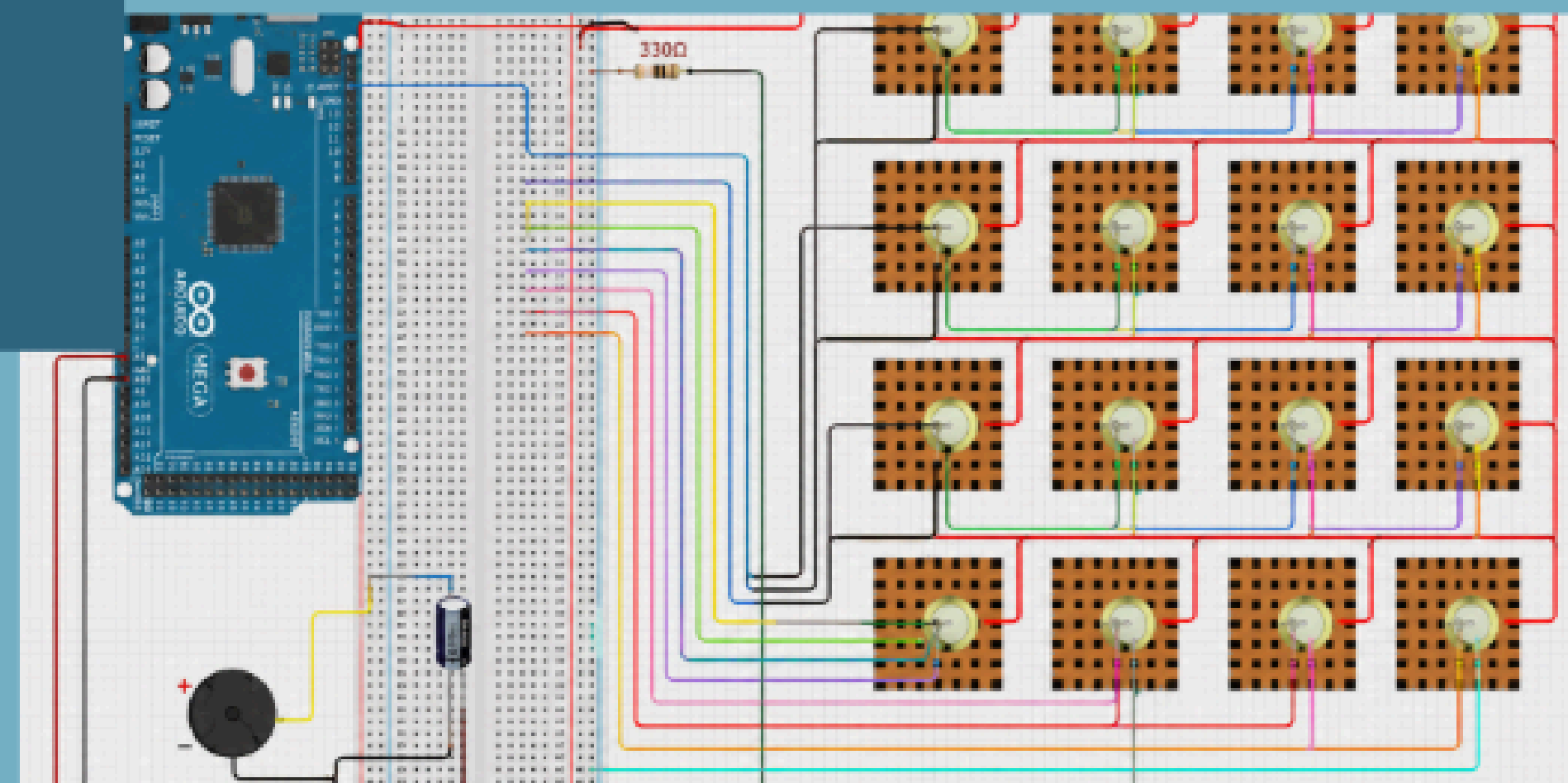
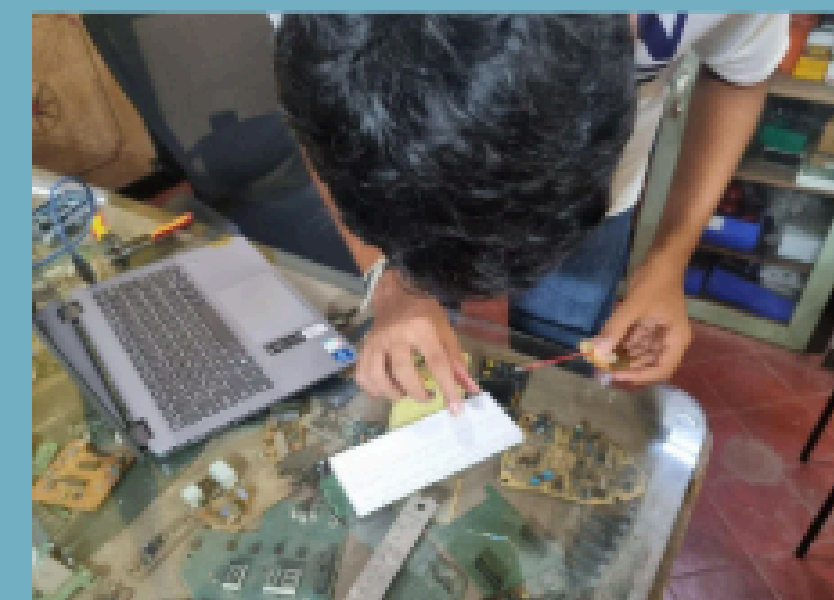
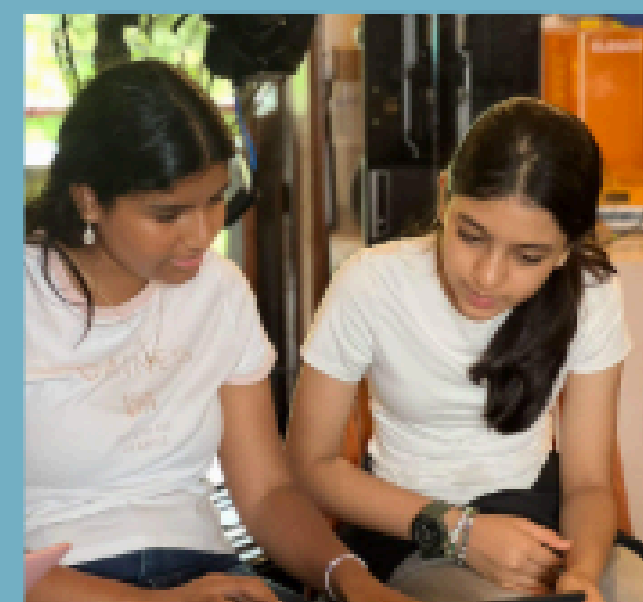


Where it All Started:

- First narrowed down to sports tech
- Had common interest
- Felt the need that sports players needed practice outside the court
- Wanted to make it engaging

Project Vector was created!

Long calls including discussions about possible project ideas and also combining two ideas which included creating a tennis game along with training a few skills



Technical Details

Components Utilized:

Arduino Mega 2560
Piezoelectric Sensors
NeoPixel LED Rings
Foam Pads
Acrylic Material
PCB
Plywood Back Panel
Power Supply

Full maker stack :designing laser-cut DXF files in CAD and 3D printing custom T-pin components, to writing Arduino firmware for 12 individually controlled WS2812B LED rings and piezoelectric impact sensors.

Every component was tested through deliberate trial and error, with soldering, circuit debugging, and threshold calibration done by hand across multiple iterations.



Challenges Faced:

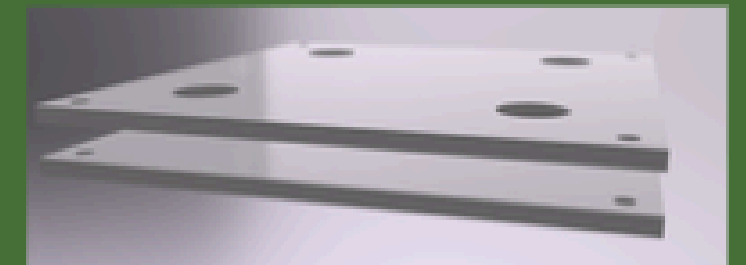
SIZE OF THE PROJECT AND SCALABILITY

A major setback in our project was that we originally planned for 16 pads with around 3 ft x 3 ft being the size. We ran into multiple issues when it came to time management and sizing limits in the laser cutting and 3D printing machine



FIXING A MEASUREMENT BY CONSIDERING MANY FACTORS

After consideration, we decided to scale it down to 12 pads with a 2 ft x 2 ft size along with understanding the complications that would arrive in terms of time if it was a larger size.



17 cm x 17 cm

WIRES ENTANGLING AND ORGANIZING

192 LEDs and 12 piezos meant a lot and lot of soldering which also means a lot of wires. In total we had over 120+ wires which we had to connect to the PCB which really made the whole setup look shabby



SIMPLE SOLUTION FOR EFFICIENCY

Anool sir suggested that we not waste time and simply elevate each pad and let all the wires pass through from beneath



Project Vector Unleashed



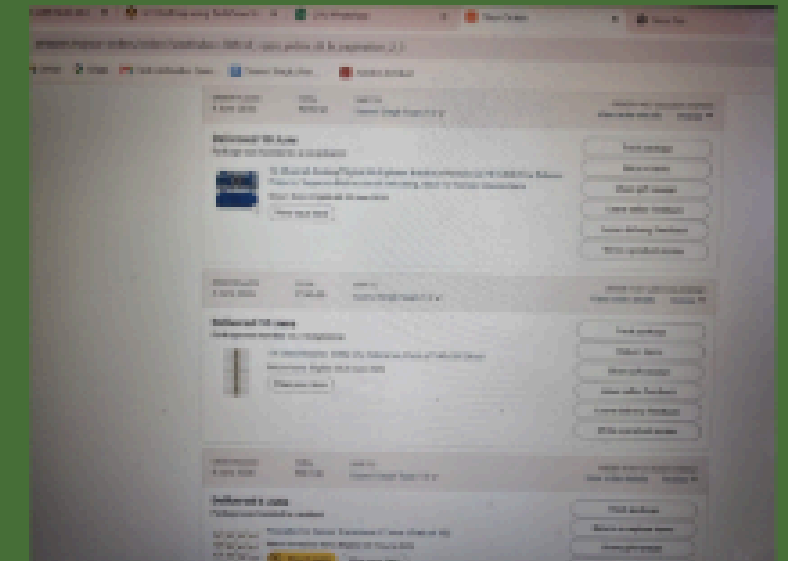
Project Vector is a wall-mounted smart target system made up of 12 interactive pads arranged in a grid, each embedded with a piezoelectric sensor that detects the impact of a tennis ball and a WS2812B LED ring that lights up to indicate the target

HOW IT WORKS:

Players see a random pad light up and have a fixed time to strike it with a tennis ball. The piezoelectric sensor underneath detects the impact, the Arduino registers a hit or miss, and the system instantly moves to a new randomized pad

BUILT WITH :

2 piezoelectric sensors, 12 WS2812B RGB LED rings, an Arduino Mega 2560 controlling every input and output directly, a 5V regulated power supply, and a laser-cut wooden frame housing each pad — all wired and soldered by hand.



WHY IT MATTERS:

Sports-tech training tools like this typically cost lakhs of rupees and live only in professional academies. Project Vector recreates that same reaction-and-accuracy training experience using affordable, open electronics — built entirely by us, from soldering to code.



Thank You